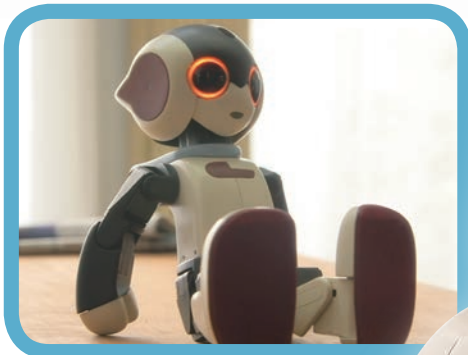
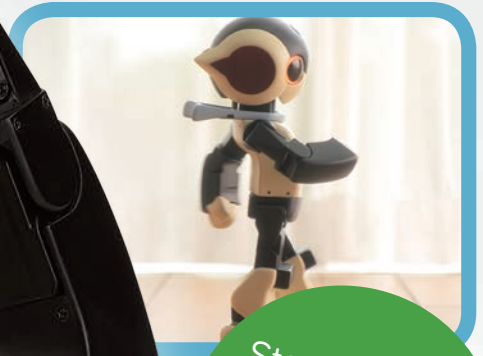
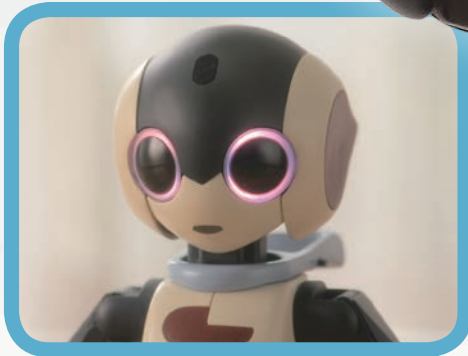


# Build your own Robi

Pack 09



Stages 31-34:  
Continue Robi's  
right leg, and  
prepare his  
'shoes'

# Build your own **Robi** CONTENTS

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Stage 34: Robi's padded shoes p169

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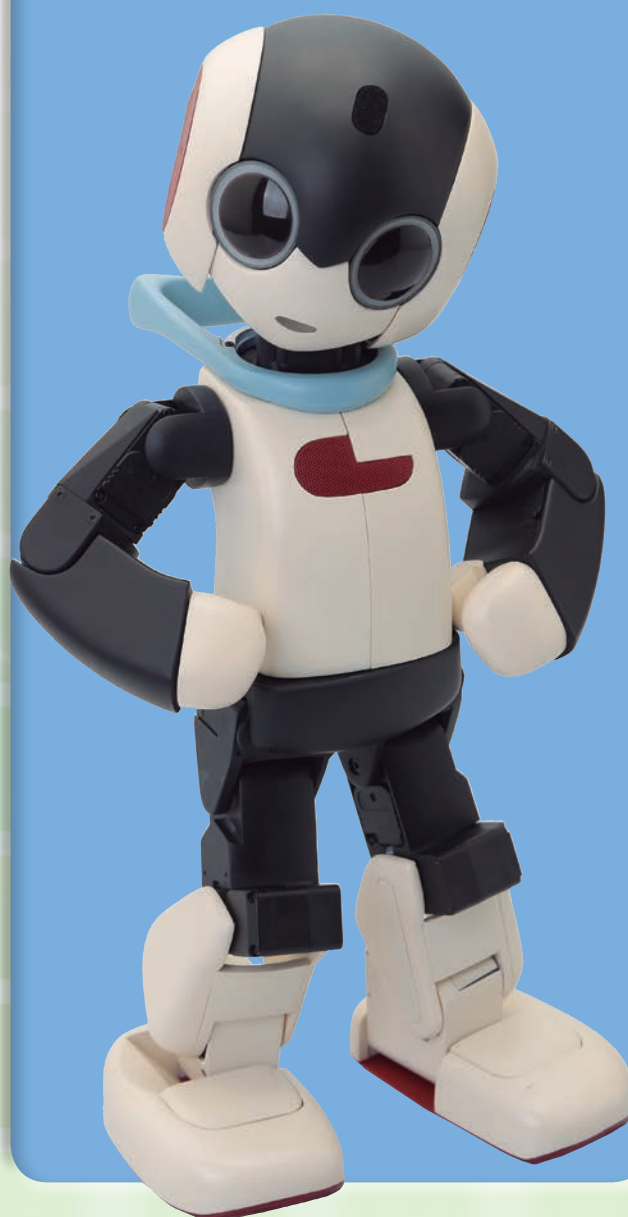
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# STAGE 31: CONTINUE BUILDING UP ROBI'S RIGHT LEG

Set the ID number of Robi's right thigh servo, then fit it to the right leg assembly built over the previous stages.

In this session, you will program the servo that will move Robi's right upper leg back and forth, servo ID=3. The process of setting the servo's ID number is the same as for all his other servos, but, as ever, be sure to follow

the steps exactly, as going back to repair any errors can be time-consuming. You will then fit the servo into the correspondingly shaped knee frame, which will join it to his hip in the coming stages.



1



## YOUR PARTS

1 Servo motor (ID=3)

**PARTS TO BE ASSEMBLED**

## SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Right leg assembly  
(Stage 30)*



*Servo cable  
(Stage 30)*



*Neck stand  
(Stage 8)*

## THE SERVO AND CABLE

TIP!

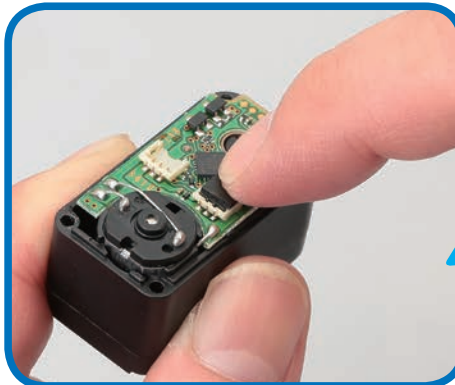
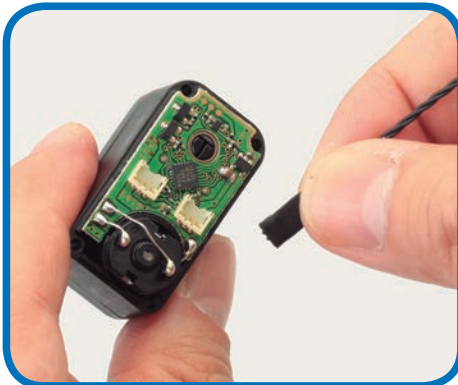


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure you hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



**1** Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

**2** Gently lift the cover away, taking care not to force it or damage the circuit board underneath it.



If you find it difficult to attach the cable with the pad in place, or if you chose not to fit it with pads in the previous stage, it is possible to apply the pad after fitting the connector. Take extra care when handling the unprotected connector. Also be careful not to press down on or obstruct the circled components on the servo's circuit board.

**3** Take the servo cable prepared in Stage 30 and fit one of its connectors onto one of the two sockets, as shown (it does not matter which one). As ever, take great care when handling exposed electronic components.

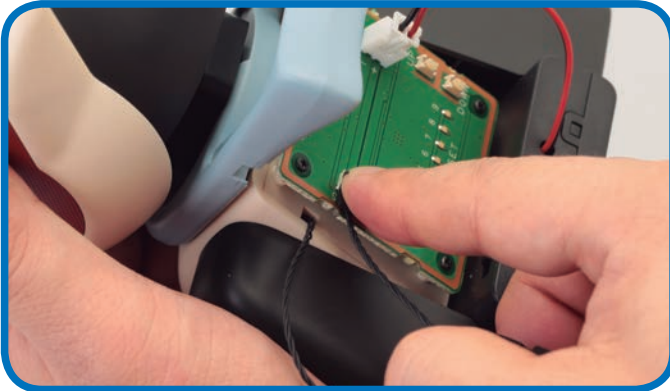


**4** Take the servo cover removed in Step 1 and fit it back onto the open servo, feeding the free tip of the servo cable through its central hole as you do.

**5** Press the cover securely back into place.

**6** Lightly tighten two of the screws removed in Step 1 into opposite corners of the servo cover. Keep all four of the screws used in this stage, as you will need them again.

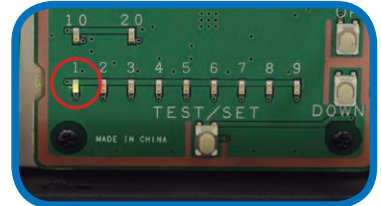
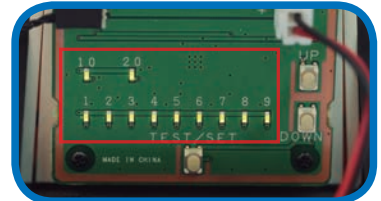
## TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board, and attach the servo cable connected to the servo in Step 3.

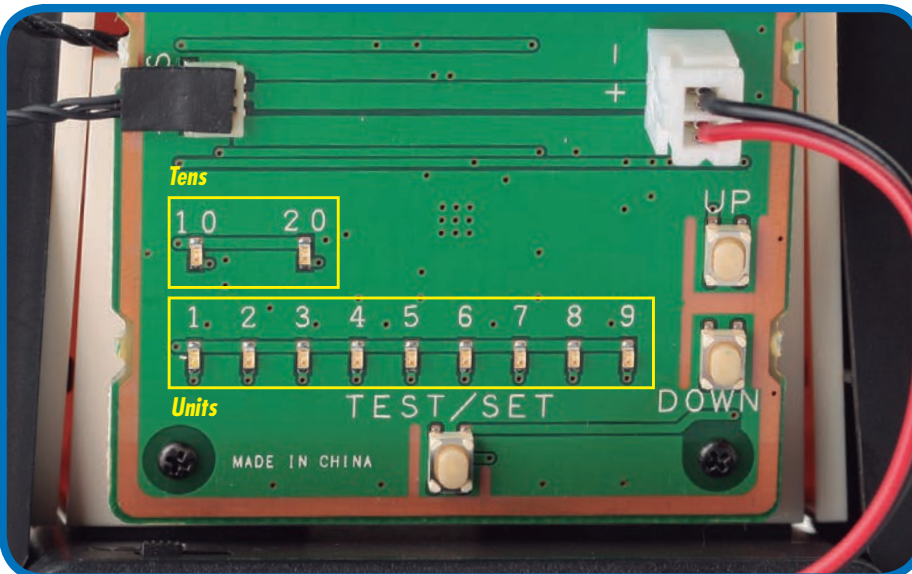


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

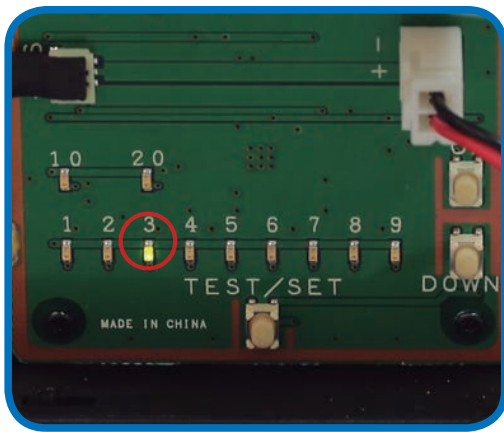


- 9** To test the servo, press the TEST/SET switch: the servo shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

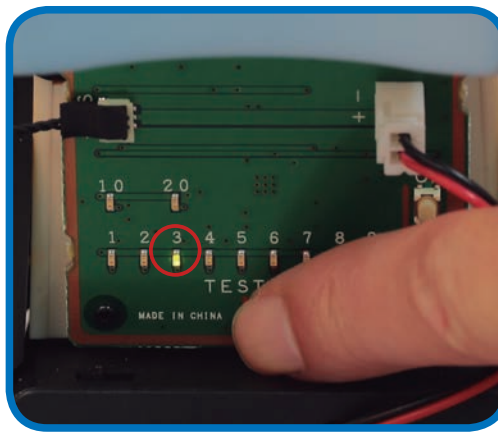
## SETTING THE ID NUMBER



- 10** Re-familiarise yourself with the lights and switches on the test board. These are used to set the ID numbers of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 3, as you will do in this stage, you press UP twice. This will light up the 3 LED alone. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, set it by pressing and holding the TEST/SET switch.



- 11** To set the ID number of Robi's right thigh servo to 3, press the UP button twice. Make sure that only the 3 LED is lit, using the UP and DOWN buttons to correct the number if necessary.



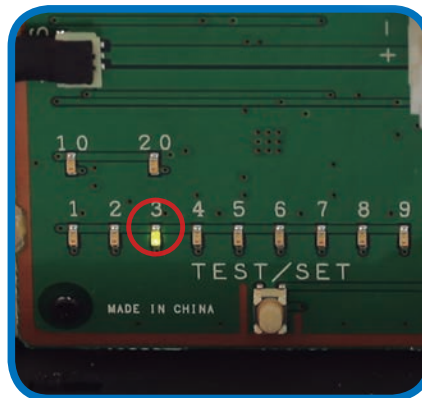
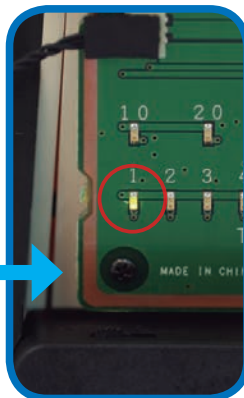
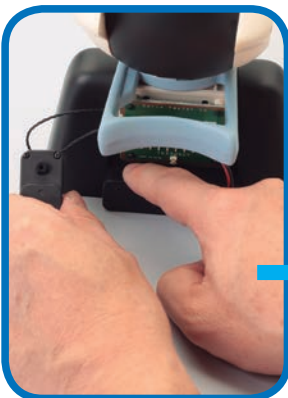
- 12** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 3 LED should flash a few times, then remain lit.



**ID=3**

## CHECKING THE ID NUMBER

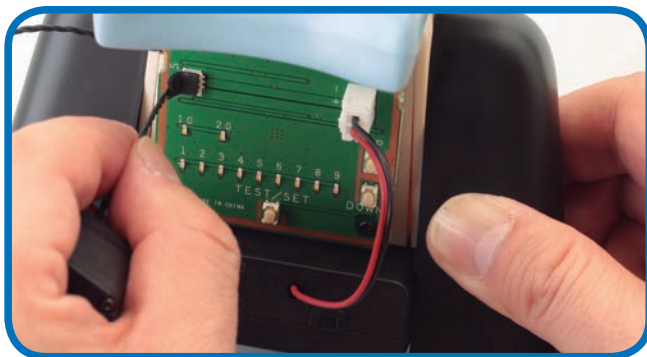
HELP!



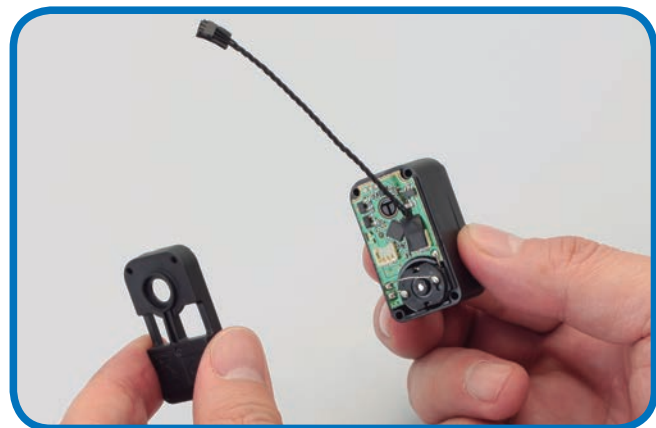
- 13** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.

- 14** To check the ID number, press TEST/SET. Now the 3 LED should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.



- 15** Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to fit into Robi's leg.



- 16** Remove the screws fitted in Step 6 and take the cover off. You will not be using the cover again.

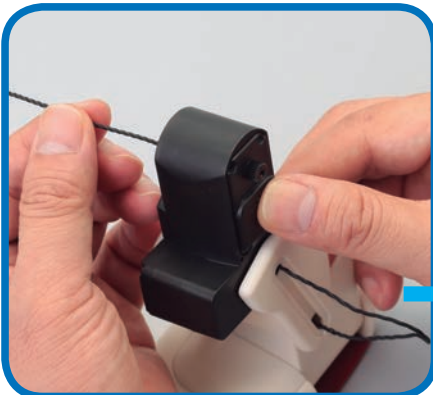
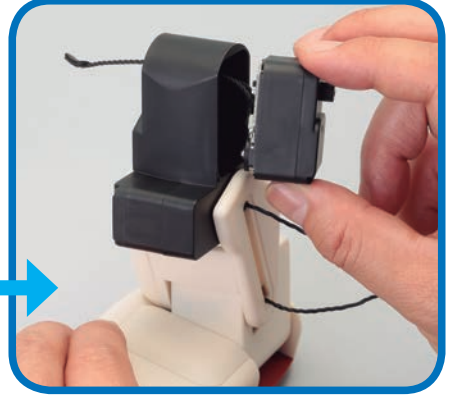
## FITTING THE THIGH SERVO



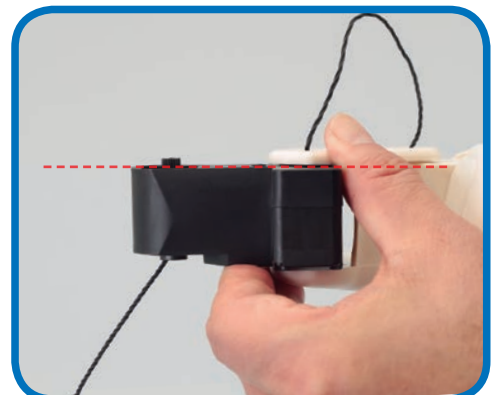
**17** Line up the thigh servo, prepared in the previous stages, to the right leg assembly. Then, very carefully, press the connector of the cable running out from the knee servo to the free socket on the thigh servo, as shown.



**18** Now feed the free tip of the thigh servo cable through the circled hole in the knee frame, from the inside of the part. Then move the servo into the correspondingly shaped frame, making absolutely sure that you do not trap the servo cables as you do so.



**19** Press the servo into the frame, so that it sits flush with the edges of the part. Again, make sure neither of the servo cables is caught, crushed or trapped in any way, and that they remain securely attached to their sockets.



**20** View your assembly from the side to double-check that the parts are sitting flush with each other.



**21** Use the four screws that originally held the thigh servo cover in place to secure the parts, as shown.



**NOTE:** As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

### Assembled right leg

*The servo that will operate Robi's right thigh movement is now in place, ready to be joined to his hip in the following stages.*

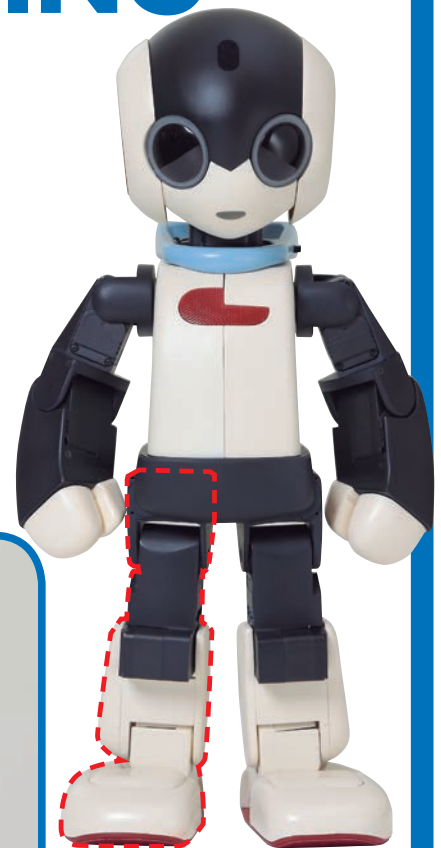


# STAGE 32: MOUNTING THE HIP JOINT

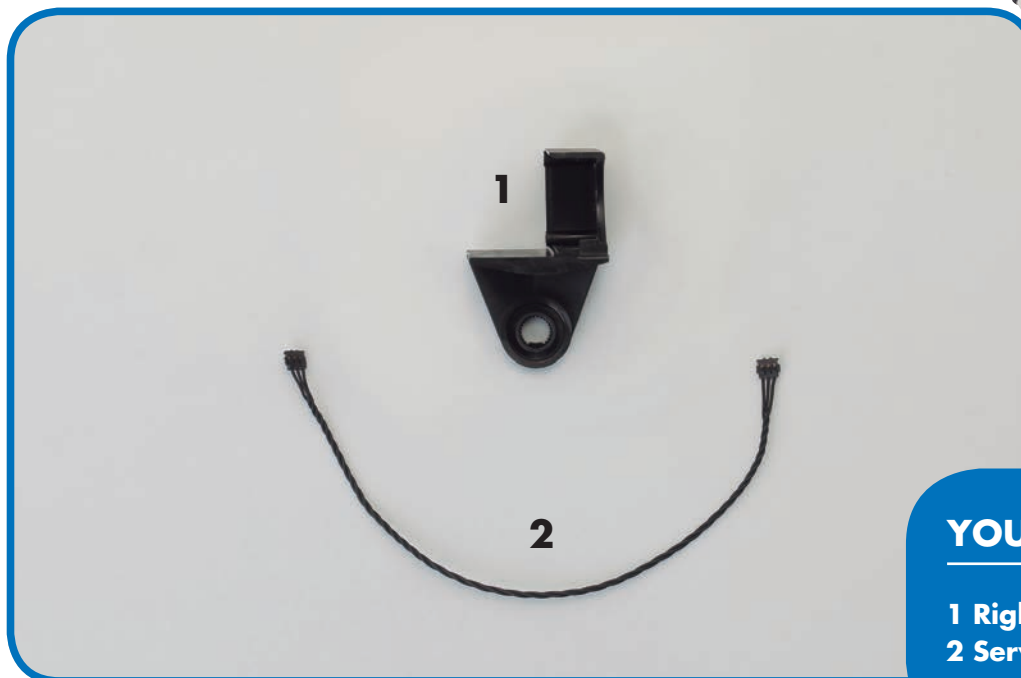
In this session, you will continue the foot-to-hip assembly of Robi's right leg. Start by mounting the inner hip frame that will join Robi's right thigh to his hip, before fitting protective pads to another servo cable.

This is a fairly straightforward stage, but it will see you working with D-cut parts again, so as ever, great care must be taken not to damage these. Remember that the parts are designed

to fit easily and neatly together – you will never need to force them into position! If they are not joining easily, simply separate them, re-align them, then try again.



**PARTS TO BE ASSEMBLED**



## YOUR PARTS

- 1 Right inner hip frame
- 2 Servo cable (135mm)

## SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



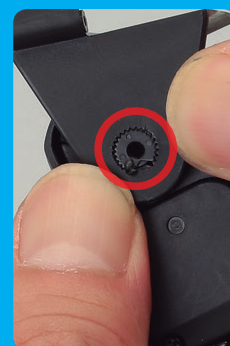
*Right foot and lower leg assembly (Stage 31)*



*Protective pads (Stage 3)*

## MOUNTING THE INNER HIP FRAME

**TIP!**



Be sure to set the frame's hole onto the servo's shaft with their D-cut profiles perfectly aligned! Never force the parts, as this can result in permanent damage. See the Close Up box in Stage 28 for more details.

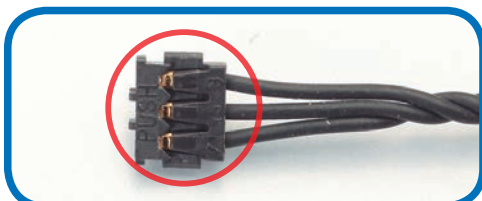
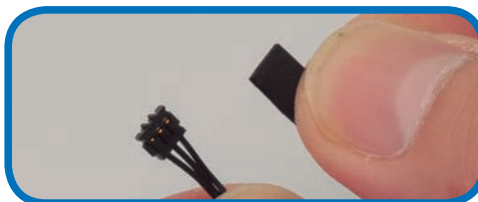
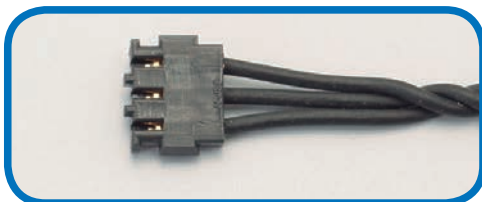
**1** Line up the inner hip frame to the right leg assembly built over the previous stages, oriented as shown. The D-cut hole in the frame will fit over the correspondingly shaped shaft of servo ID=3, the thigh servo.

**2** Making absolutely sure the shaft is sitting correctly within the D-cut hole, press the parts together.

## SEALING THE SERVO CABLE

### Assembled right leg

*Working up from his foot, you have now added the first of Robi's hip frames, which will join his right thigh to his right hip servo on completion.*



**3** As you have done previously, identify the side of this stage's 135mm-long servo cable connector that has PUSH marked on it (circled).

**4** Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

**TIP!**

During the build, you may have found it easier to attach the servo connectors to the servos' sockets without the protective pads pre-fitted. If this is the case, you can skip the 'Sealing the servo cable' steps above and fit them later. However, the pads are designed for the connectors' protection, so if you handle them without these in place, take extra care.



# USEFUL TOOLS AND TIPS



The following are a few tips to help you to build your Robi easily and efficiently.

## Magnetising your screwdriver

Some screwdrivers have magnetic tips, which attract steel screws to help reduce the risk of losing them during a build. The special screwdriver supplied in Pack 1 is not magnetised, because the screws used for the Robi build are so small that if it were, it might pick up several at a time, which could cause difficulty. However, if you feel it would be useful to have it magnetised for a particular step, it is easy to make it temporarily magnetic.



To enable your screwdriver to pick up steel screws, all you need is a magnet. A household fridge magnet is ideal. Simply stroke this downwards along the black tip of your screwdriver, repeating the stroke several times in one direction only. After a number of strokes, your driver will become magnetic!

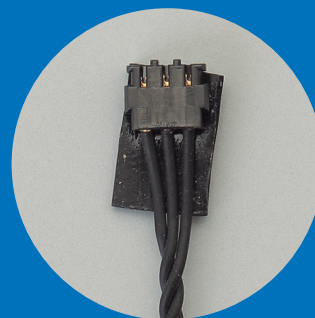


As well as helping you pick screws up, magnetising your screwdriver will help set screws in harder-to-reach places.

## Servo cables, connectors and protective pads

Every servo cable connector must be fitted with a protective pad once fitted to its servo, as otherwise the small and delicate parts can get damaged during the build and impede Robi's operation when completed. However, while we have advised that it is safer to fit these pads before joining the connectors to their sockets, you may have found it easier to do so once they are in place. If you choose to do this, take extra care when handling the parts. It is also vital that you never force the connectors when fitting them: they are designed to click into place neatly, so if they do not, simply adjust their position until they do.

*It is also important that the pads be fitted perfectly straight to their connectors, or they will get in the way once fitted inside the servos themselves.*



## Storing spare screws

Whenever you have received screws with a stage's parts, you have been given one of each type as a spare. These will be useful in case you lose a screw during an assembly session, if the head of a screw gets damaged, making it hard to turn, or for maintenance of your Robi once completed. In any case, it will be useful to store all your spare screws in an orderly manner, and for this a compartmentalised hobby box will be ideal. These are available at modelling shops, or even homeware shops, and will help you separate your screws by size and type.



*There are many types of box that will be suitable – pill boxes, normally used to store medication, are particularly well suited to the task.*



# STAGE 33: PREPARE ROBI'S RIGHT HIP SERVO

In this assembly, you will test and program the servo that will operate the side-to-side movement of Robi's right hip, ready to be joined up to the rest of his leg assembly in coming stages.

This stage sees you working with servo motors again. The servo supplied with this stage is ID=2, as listed on the diagram in Stage 8, and will produce the side-to-side motion

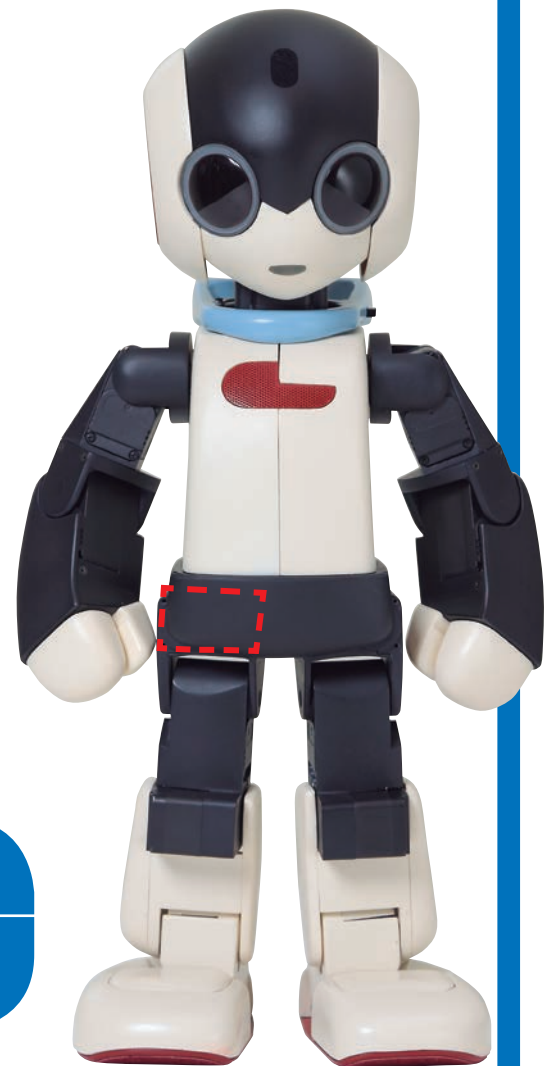
for Robi's right hip. Once it is set, the servo will be fitted to the hip and knee frames, enabling his leg to move in a natural way. As always, take care when handling the electronic components!



**PARTS TO BE  
ASSEMBLED**

## YOUR PARTS

1 Servo motor (ID=2)



## SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



*Servo cable (Stage 32)*

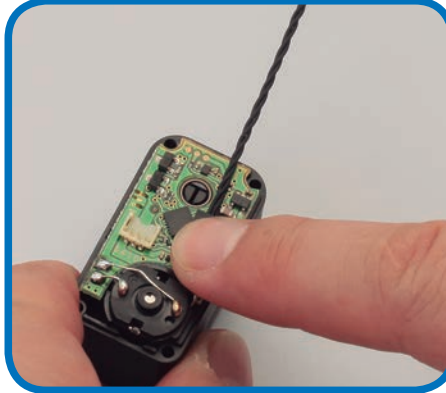


*Neck stand (Stage 8)*

## TESTING THE SERVO



- 1** Remove the cover from the servo, retaining both it and the four screws.



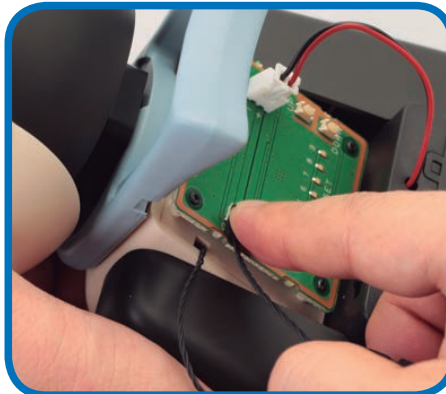
- 2** Fit one of the connectors of the 135mm-long servo cable prepared in Stage 32 into one of the sockets.



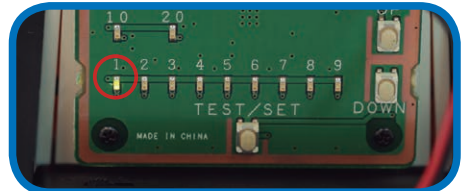
- 3** Replace the cover, feeding the servo cable through the circular hole in its centre, as shown.



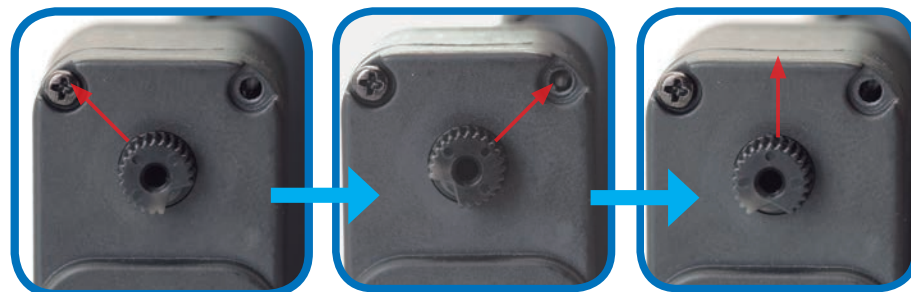
- 4** Secure the cover with all four of the screws removed in Step 1.



- 5** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable that you connected to the servo in Step 2.



- 6** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

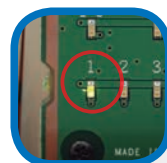
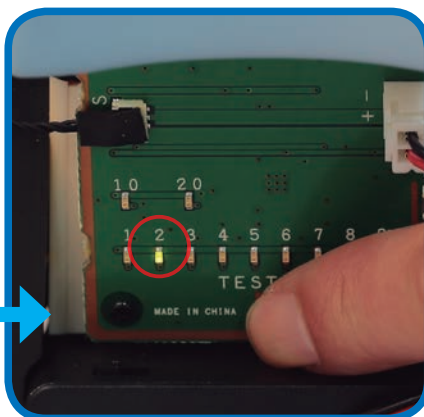
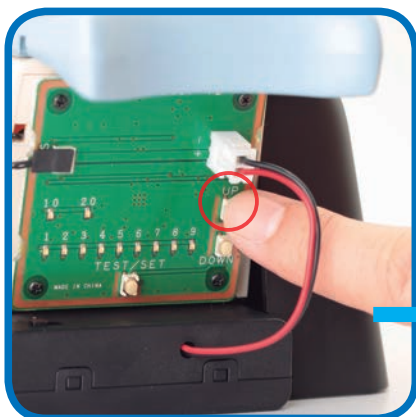


- 7** To test the servo, press the TEST/SET switch: the servo shaft should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. The next step is to set the servo's ID number, so keep the power on.

### TIP!

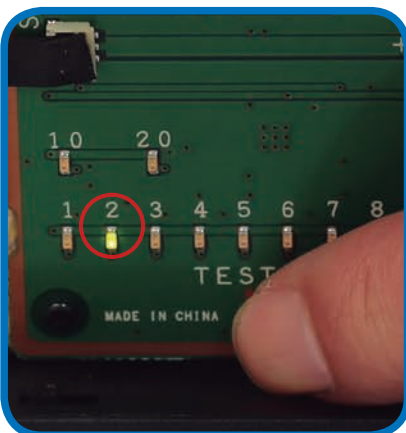
If the test does not run when you press the TEST/SET button, or if the LEDs flash rapidly, you may have a faulty connection. Check that the connector is fitted correctly into the test board's socket, and if it is, carefully unscrew the cover of the servo (replaced in Step 4) and check the connection there. For more info on this, refer to the HELP box in Stage 8.

## SETTING THE ID NUMBER



**8** The servo that will move Robi's right hip is ID=2, so to set the ID number to 2, press the UP button just once. Complete the setting by holding the TEST/SET button down for a few seconds. The 2 light should flash, then stay lit, confirming the ID number.

**9** Turn the power OFF, then back ON. The 1 LED should light up.



**10** To check the ID number, press TEST/SET. Now only the 2 LED should light up, as the servo shaft rotates as in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

Once you are absolutely certain that the servo is set correctly, turn the power OFF and carefully remove the servo cable from the test board. Pull the cable away gently, directly upwards from the socket (never pull at an angle). This is especially important if you have not yet applied a protective pad to the connector.

**11**



## Programmed servo ID=2

*The servo that will operate Robi's right hip is now programmed with its ID number, and is ready to be fitted to the rest of the right leg assembly in the coming stages.*



*It will be helpful to store this stage's servo in a sealed plastic bag with the stage number written on it, along with the ID number.*

# STAGE 34: ROBI'S PADDED SHOES

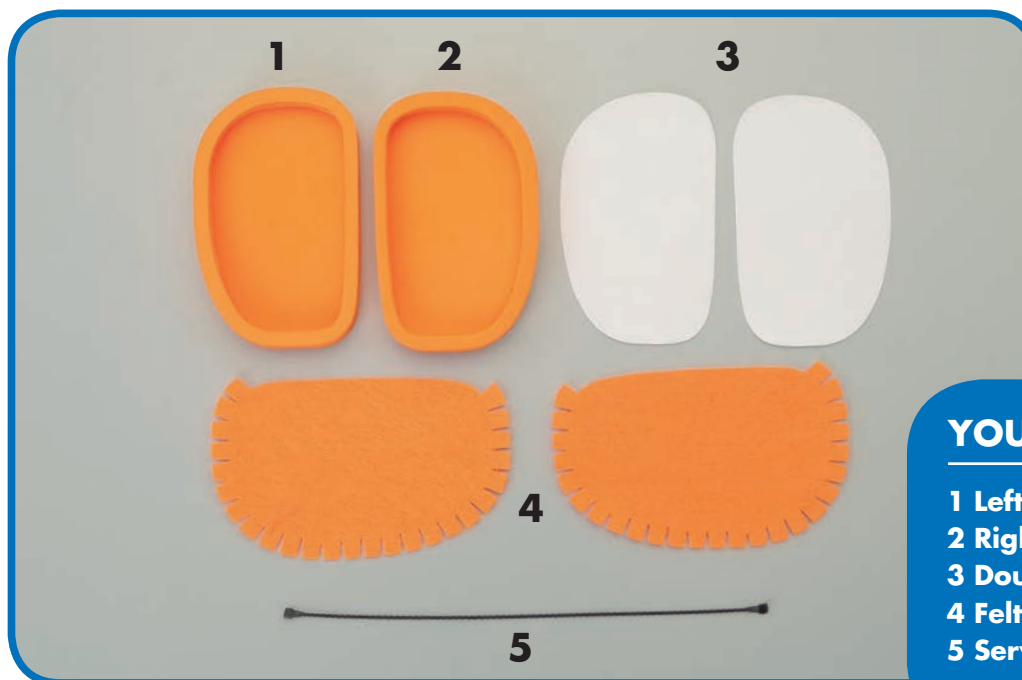
Robi comes with two padded, felt-soled shoes, with which he can 'clean' your tabletop or floor with a special walk. In this session, you will stick the felt pads to the soles of these shoes, ready for use when Robi is operational.

When complete, Robi will be able to use specially designed felt-soled shoes to 'clean' your floor or tabletop with a pre-programmed walk. In this stage, you will simply add the felt soles

to these shoes, which you will do by using two double-sided adhesive pads. The pads are identical, and can go on either shoe, but be especially careful to set the felt soles on straight.



**PARTS TO BE ASSEMBLED**



## YOUR PARTS

- 1 Left shoe
- 2 Right shoe
- 3 Double-sided adhesive pads
- 4 Felt soles
- 5 Servo cable (200mm)

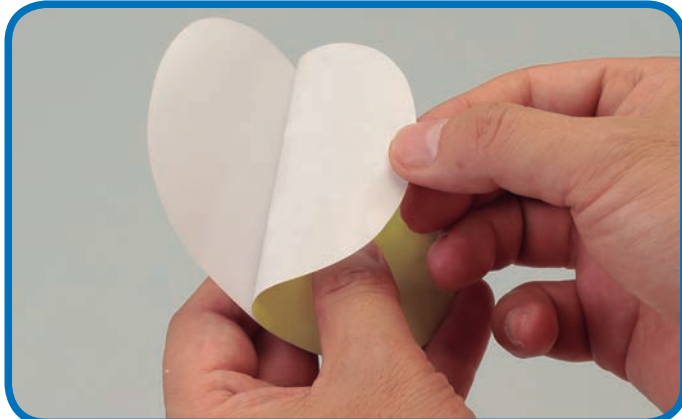
## SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



*Protective pads  
(Stage 3)*

## FITTING THE FELT SOLES



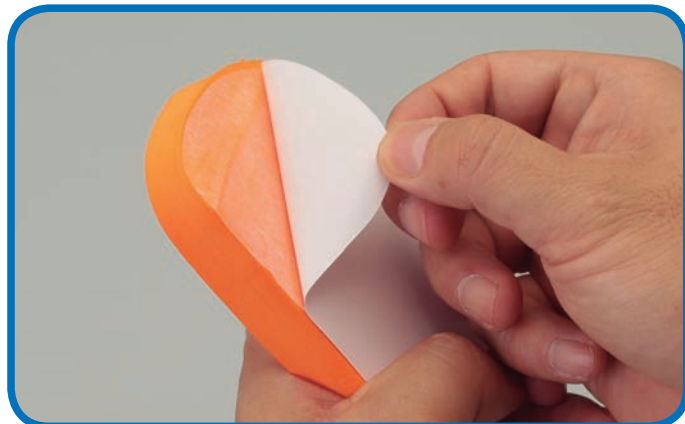
- 1** Take one of the double-sided adhesive pads, and peel away the protective film on one side of it.



- 2** Line up the sticky side of the pad with the sole of the shoe that it matches. Carefully fit it to the shoe, making sure that it's set straight and starting from the top edge.



- 3** Work down the pad, pressing evenly to stick it to the underside of the shoe.



- 4** Now peel away the second protective film to reveal the adhesive surface beneath it. Do not touch the adhesive surface, which might make it less effective.



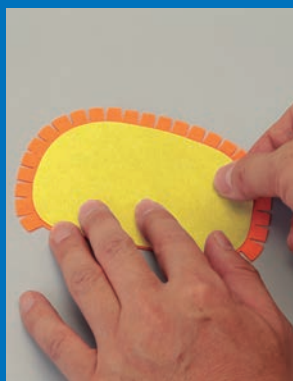
- 5** Line the shoe up with its felt sole – the straight inside edge of the shoe will match that of the sole, as shown by the dashed red outline above.



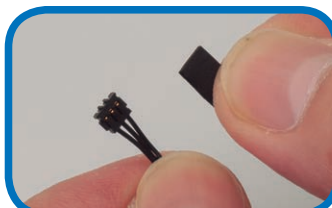
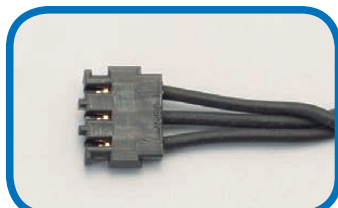
- 6** Press the shoe and the sole together to secure them. Then repeat these six steps to fit the felt sole to Robi's other shoe (see also the Close-Up box on the next page).



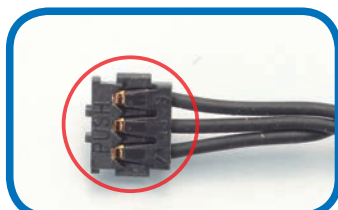
*When fitting the second shoe, if you accidentally remove the wrong side of the double-sided adhesive pad first, meaning that it won't fit to the underside of the shoe, simply stick it on the sole first. Fit the pad to the top of the felt sole, and then stick the underside of the shoe onto it, as shown.*



## SEALING THE SERVO CABLE



It is advisable to fit the servo cable's protective pads at this stage, but you can wait until the cable is used in the build, if you prefer. In that case, simply store the cable away safely until you use it, remembering to apply the pad then.



- 7** As you have done previously, apply a protective pad to the side of this stage's longer, 200mm servo cable with PUSH marked on it (circled). Repeat for both ends of the cable.

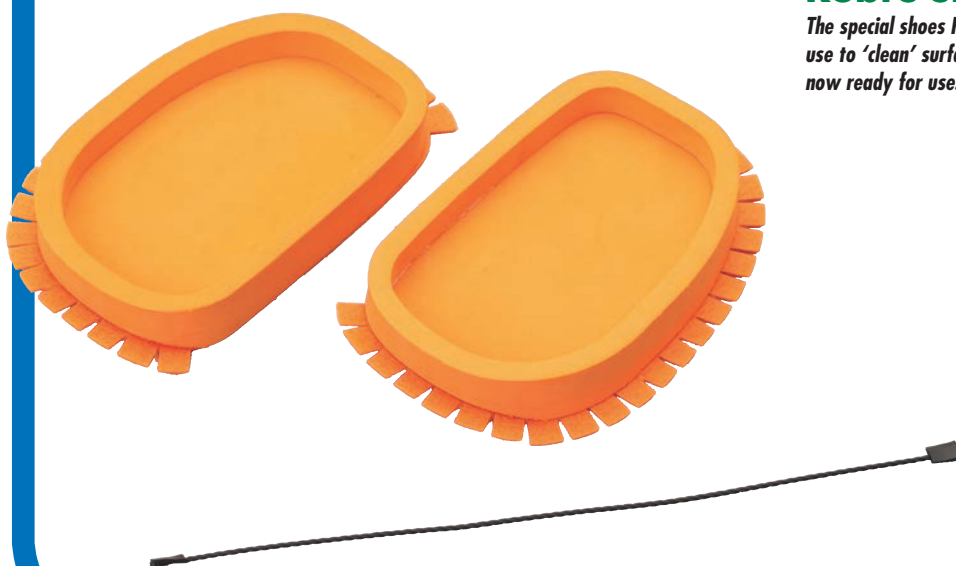
## Robi's shoes

*The special shoes Robi will use to 'clean' surfaces are now ready for use.*

**TIP!**



You will not be using these shoes until Robi is complete, so store them away safely until then. Keep them in a dust-free environment and out of direct sunlight. Note that Robi will only wear these shoes when in 'cleaning' mode, and not during ordinary operation. More information will be given at the conclusion of the series.



# COMING IN PACK 10

Your next **FOUR** complete Stages, as Robi starts to come to life!



## Stage 35

Prepare Robi's first left foot servo

### PARTS PROVIDED

The servo that will move Robi's left foot.



## Stage 36

Robi's left foot cover

### PARTS PROVIDED

The left foot cover, ankle side panel and ankle joint.



## Stage 37

Robi's left ankle frame

### PARTS PROVIDED

The frame that will join Robi's foot and ankle servos, and a servo cable.



## Stage 38

Robi's battery pack

### PARTS PROVIDED

The battery pack that will power Robi on completion.